6. [Maximum mark: 5]

Consider the infinite series

$$\sum_{k=0}^{\infty}\left(\frac{53}{1000}\right)\left(\frac{1}{100}\right)^{k}.$$

(a) Find the exact value of S_{∞}. [3]

Let $0.2\overline{53}$ represent the repeating decimal $0.2535353...$.

(b) Use your answer from part (a) to express $0.2\overline{53}$ as a fraction such that the numerator and denominator have no common factors. [2]